Trigonometry

ANSWER KEY

Right triangle trigonometric ratios

1 In a right triangle, angle θ has an opposite side of 8 and hypotenuse 17. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.
The adjacent side: $\sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$. $\sin\theta = \frac{8}{17}$, $\cos\theta = \frac{15}{17}$, $\tan\theta = \frac{8}{15}$.

2 A ladder 15 feet long leans against a wall, making a 70° angle with the ground. Find the height reached on the wall, to the nearest tenth.
Height $= 15\sin(70^\circ) \approx 15(0.9397) \approx 14.1$ feet.

The unit circle and radian measure

3 Convert 150° to radians, in terms of π.
$150^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{6}$.

4 Convert $\frac{3\pi}{4}$ radians to degrees.
$\frac{3\pi}{4} \times \frac{180^\circ}{\pi} = 135^\circ$.

5 Find the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$, using the unit circle.
$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$. $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.

Basic trigonometric identities

6 Use the Pythagorean identity to find $\cos\theta$, given that $\sin\theta = \frac{3}{5}$ and θ is in the first quadrant.
$\sin^2\theta + \cos^2\theta = 1$, so $\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Since θ is in the first quadrant, $\cos\theta$ is positive: $\cos\theta = \frac{4}{5}$.

7 Simplify $\frac{\sin\theta}{\cos\theta} \cdot \cos\theta$.
$\frac{\sin\theta}{\cos\theta} \cdot \cos\theta = \sin\theta$ (the $\cos\theta$ terms cancel).

Graphs of sine and cosine functions

8 State the amplitude and period of $y = 3\sin(2x)$.
Amplitude: $|3| = 3$. Period: $\frac{2\pi}{2} = \pi$.

9 State the amplitude, period, and vertical shift of $y = 4\cos(x) + 2$.
Amplitude: 4. Period: 2π (unchanged, since there is no coefficient on x inside the cosine). Vertical shift: up 2 units.

Law of Sines

10 In triangle ABC, $A = 40^\circ$, $B = 65^\circ$, and $a = 10$. Find side b, to the nearest tenth, using the Law of Sines.

Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$. $\frac{10}{\sin 40^\circ} = \frac{b}{\sin 65^\circ}$. $b = \frac{10\sin 65^\circ}{\sin 40^\circ} \approx \frac{10(0.9063)}{0.6428} \approx 14.1$.

Law of Cosines

11 In triangle ABC, $a = 7$, $b = 9$, and $C = 60^\circ$. Find side c, using the Law of Cosines.

$c^2 = a^2 + b^2 - 2ab\cos C = 49 + 81 - 2(7)(9)\cos 60^\circ = 130 - 126(0.5) = 130 - 63 = 67$.
$c = \sqrt{67} \approx 8.2$.

Trigonometric equations

12 Solve $2\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$.

$\sin\theta = \frac{1}{2}$. Solutions in $[0^\circ, 360^\circ)$: $\theta = 30^\circ$ or $\theta = 150^\circ$ (both quadrants where sine is positive).

Applications of trigonometry to real-world problems

13 This question has two parts. A surveyor stands 200 feet from the base of a tower and measures the angle of elevation to the top as 38°. (a) Find the height of the tower, to the nearest foot. (b) If the surveyor moves 50 feet closer to the tower, find the new angle of elevation, to the nearest degree, using the height found in part (a).

(a) Height $= 200\tan(38^\circ) \approx 200(0.7813) \approx 156$ feet. (b) New distance: $200 - 50 = 150$ feet. New angle: $\theta = \arctan\left(\frac{156}{150}\right) = \arctan(1.04) \approx 46^\circ$.

Reciprocal trigonometric functions

14 If $\sin\theta = \frac{5}{13}$, find $\csc\theta$.

$\csc\theta = \frac{1}{\sin\theta} = \frac{13}{5}$.

15 If $\cos\theta = \frac{3}{5}$, find $\sec\theta$ and $\tan\theta$ (assuming θ is in the first quadrant).

$\sec\theta = \frac{1}{\cos\theta} = \frac{5}{3}$. Since $\sin^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$, $\sin\theta = \frac{4}{5}$ (positive in the first quadrant).
$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{4/5}{3/5} = \frac{4}{3}$.

Angle of elevation and depression

16 From the top of a 100-foot cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

The angle of depression from the top equals the angle of elevation from the boat. Using the tangent ratio: $\tan(25^\circ) = \frac{100}{d}$, so $d = \frac{100}{\tan(25^\circ)} \approx \frac{100}{0.4663} \approx 214$ feet.

Solving trigonometric equations over a restricted domain

17 Solve $2\cos\theta + 1 = 0$ for $0^\circ \le \theta < 360^\circ$.

$\cos\theta = -\frac{1}{2}$. Solutions in $[0^\circ, 360^\circ)$: $\theta = 120^\circ$ or $\theta = 240^\circ$ (both quadrants where cosine is negative).

A multi-part triangle problem combining Law of Sines and Law of Cosines

18 This question has two parts. Triangle ABC has $a = 8$, $b = 11$, and $C = 50^\circ$. (a) Use the Law of Cosines to find side c, to the nearest tenth. (b) Use the Law of Sines to find angle A, to the nearest degree.

(a) $c^2 = 8^2 + 11^2 - 2(8)(11)\cos 50^\circ = 64 + 121 - 176(0.6428) \approx 185 - 113.1 \approx 71.9$. $c \approx \sqrt{71.9} \approx 8.5$. (b) $\frac{a}{\sin A} = \frac{c}{\sin C}$: $\frac{8}{\sin A} = \frac{8.5}{\sin 50^\circ}$. $\sin A = \frac{8\sin 50^\circ}{8.5} \approx \frac{8(0.766)}{8.5} \approx 0.721$, so $A \approx 46^\circ$.

Cofunction and negative angle identities

19 Use the cofunction identity to find $\cos(30^\circ)$ given that $\sin(60^\circ) = \frac{\sqrt{3}}{2}$.

The cofunction identity states $\cos\theta = \sin(90^\circ - \theta)$, so $\cos(30^\circ) = \sin(90^\circ - 30^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2}$.

20 Use the fact that sine is an odd function to find $\sin(-30^\circ)$, given $\sin(30^\circ) = \frac{1}{2}$.

Since sine is odd, $\sin(-\theta) = -\sin(\theta)$: $\sin(-30^\circ) = -\sin(30^\circ) = -\frac{1}{2}$.